Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 22)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A pesticide was applied to a field of oilseed rape. Table 1 shows the concentration of the pesticide in organisms sampled from a pond next to the field.

Table 1

Organism	Trophic level	Pesticide concentration / mg kg^{-1}
Pond water	-	0.002
Phytoplankton	1	0.04
Water fleas (Daphnia)	2	0.5
Small fish	3	4.2
Heron	4	38.0

0 1 . 1 Calculate how many times more concentrated the pesticide is in the heron than in the pond water. Give your answer to two significant figures.

[2 marks]

Answer ☐ times

0 1 . 2 Name the process shown by the data in Table 1.

[1 mark]

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0 1 . 3 Explain the **two** properties of the pesticide that cause the pattern shown in Table 1.

[4 marks]

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0 1 . 4 The pesticide was probably an organochlorine. Suggest why the use of organochlorines is now banned or heavily restricted in most countries, even though DDT is still used to control malaria mosquitoes in some houses.

[3 marks]

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0 2 . 1 Complete Table 1 to show a suitable sampling technique for each group of organisms and one limitation of the technique.

[6 marks]

Table 1

Organisms	Technique	One limitation	
Invertebrates active on the ground surface at night			
Invertebrates living in leaf litter			
Invertebrates on tree branches			

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0 2 . 2 Explain why a light trap cannot be used to prove that a moth species is absent from a site.

[2 marks]

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0 2 . 3 Explain how a motor-driven suction sampler can give a population density, whereas a pooter cannot.

[2 marks]

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0 3 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☒

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

0 3 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 3 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

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0 4 In 2020 the tanker MV Wakashio ran aground on a coral reef off Mauritius and leaked about 1000 tonnes of fuel oil into a shallow lagoon near mangroves and a marine park.

0 4 . 1 Explain why inflatable booms were deployed immediately, and give **one** limitation of booms.

[3 marks]

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0 4 . 2 Describe how skimmers and absorbent materials remove oil, and state how the collected oil is disposed of.

[3 marks]

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0 4 . 3 Explain why dispersants were **not** used in the lagoon.

[2 marks]

0 4 . 4 Explain how bioremediation could help remove the oil that remained on the mangrove sediments, and suggest how it could be accelerated.

[2 marks]

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0 5 . 1 Describe how sulfur can be removed from coal **before** combustion.

[2 marks]

0 5 . 2 Describe the process of dry flue gas desulfurisation and explain how the product can be used.

[3 marks]

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0 5 . 3 In wet flue gas desulfurisation the flue gases are bubbled through sodium sulfite solution. Explain **one** advantage of this method over dry FGD.

[2 marks]

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0 5 . 4 Explain how each of the following reduces NOx emissions.

[3 marks]

Fluidised bed combustion .....

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Catalytic converters .....

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Urea sprays .....

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0 6 . 1 Explain how each of the following building design features reduces energy use.

[6 marks]

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Orientation

Low surface area : volume ratio

Earth sheltering

0 6 . 2 Compare the embodied energy of concrete with alternative building materials.

[2 marks]

0 6 . 3 Describe how a heat exchanger allows a building to be ventilated with minimal heat loss.

[2 marks]

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0 7 . 1 Which of the following is the main reason that lead-based paint is a greater risk to children than to adults? Tick one box.

[1 mark]

A Children inhale more lead vapour ☐

B Children pick up dust / flakes and suck their fingers ☐

C Lead is more soluble in children's blood ☐

D Children spend more time near roads ☐

0 7 . 2 Explain how tri-butyl tin (TBT) in anti-fouling paint caused pollution even though only small amounts dissolved into the sea.

[3 marks]

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0 7 . 3 Iron is not toxic, yet iron-rich drainage from mine spoil heaps kills fish. Explain why, and describe how it is controlled.

[3 marks]

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0 7 . 4 Give **two** uses of lead that continue because they do not cause significant lead release into the environment.

[2 marks]

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0 7 . 5 Explain why heavy metals are described as neurotoxins.

[1 mark]

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0 8 Figure 1 shows a simplified energy budget for the Earth and its atmosphere. Values are percentages of incoming solar radiation.

Figure 1

Energy flow	Percentage of incoming solar radiation / %
Reflected by clouds, dust and atmosphere	25
Reflected by the Earth's surface	5
Absorbed by the atmosphere	20
Absorbed by the Earth's surface	50
Infrared emitted by the surface and absorbed by greenhouse gases	95
Infrared re-emitted by the atmosphere back to the surface	85
Infrared emitted to space	70

0 8 . 1 Use Figure 1 to calculate the planetary albedo of the Earth.

[1 mark]

Answer ..... %

0 8 . 2 Describe how each of the following wavelengths of electromagnetic radiation behaves in the atmosphere.

[6 marks]

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Ultraviolet (UV) .....

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Visible light .....

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Infrared (IR) .....

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0 8 . 3 Explain why the amount of infrared re-emitted to the surface (85%) is larger than the amount emitted to space (70%).

[2 marks]

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0 8 . 4 Explain the term **thermal stratification** of the atmosphere.

[1 mark]

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0 9 . 1 Explain why planting a new forest only sequesters carbon for a limited period, and how the storage can be extended.

[3 marks]

0 9 . 2 Compare pre-combustion and post-combustion methods of carbon capture.

[4 marks]

0 9 . 3 Describe **two** ways captured CO_2 can be stored and give **one** concern about long-term storage.

[3 marks]

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1 0 . 1 Evaluate the potential contribution of renewable energy resources to future energy supplies in a country such as the UK.

[9 marks]

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Extra space

1 0 . 2 State the factor by which the kinetic energy of wind increases when wind velocity doubles.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

1 1 . 2 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS